Layer 2 Instructor Required Command Checklist

Student commands and instructor-checkable outcomes from Part 2 pages 9 through 15

Purpose

This is the narrowed Layer 2 scope for the TSS2 capstone. It lists what the student is required to perform on AN-1 and AN-2 and what the instructor is required to verify. VLAN IDs and most VLAN names are seeded in the student's Binder.

The required flow is:

- Harden both switches.
- Create the five VLANs on each switch.
- Assign AN-1 access ports, trunks, and unused ports.
- Assign AN-2 access ports, trunk, and unused ports.
- Verify both switches and save both configurations.

Device hardening on both switches

```
enable
configure terminal
hostname AN-1
enable secret <Binder enable secret>
service password-encryption
no ip domain-lookup
banner motd #WARNING: DoD System#

line console 0
 password <Binder console password>
 login
 exit

line vty 0 4
 password <Binder VTY password>
 login
 exit
```

Use hostname AN-2 when configuring the second switch.

Instructor check	Required result
Identity	Hostname is AN-1 or AN-2 as appropriate.
Privileged access	Enable secret is configured; legacy enable password is not used.
Console and VTY	Each line has the Binder password and login enabled.
Security services	Password encryption is enabled and DNS lookup is disabled.
Banner	The DoD warning banner is present with the required text.

VLAN database on both switches

```
vlan <MGMT ID>
 name <Management name>
 exit
vlan <RECEP ID>
 name <Reception name>
 exit
vlan <DATA ID>
 name <Data name>
 exit
vlan <NATIVE ID>
 name <Native name>
 exit
vlan <DEAD ID>
 name Dead_Vlan
```

```
state suspend
exit
```

Instructor check	Required result
Database completeness	All five VLANs exist on both AN-1 and AN-2.
Seeded values	IDs and role names match the student's Binder exactly.
Unused VLAN	The VLAN is named Dead_Vlan and is suspended.
Independence	AN-2 has its own VLAN database; the trunk does not replicate VLAN configuration.

AN-1 switchports

Access ports

```
interface fa0/2
 switchport mode access
 switchport access vlan <MGMT ID>
 exit
interface range fa0/3 - 4
 switchport mode access
 switchport access vlan <RECEP ID>
 exit
interface fa0/5
 switchport mode access
 switchport access vlan <MGMT ID>
 exit
```

Ports	Required assignment
Fa0/2	Management
Fa0/3 - Fa0/4	Reception
Fa0/5	Management

Trunks

```
interface fa0/1
 switchport mode trunk
 switchport trunk allowed vlan <MGMT>,<RECEP>,<DATA>
 switchport trunk native vlan <NATIVE ID>
 exit
interface fa0/6
 switchport mode trunk
 switchport trunk allowed vlan <MGMT>,<RECEP>,<DATA>
 switchport trunk native vlan <NATIVE ID>
 exit
```

- Fa0/1 carries VLANs between AN-1 and AN-2.
- Fa0/6 carries VLANs from AN-1 toward CN-1.
- The native VLAN is not included in the allowed VLAN list.

Unused ports

```
interface range fa0/7 - 8
 switchport mode access
 switchport access vlan <DEAD ID>
 shutdown
 exit
```

AN-2 switchports

Access ports

```
interface fa0/2
 switchport mode access
 switchport access vlan <DATA ID>
 exit
```

```
interface fa0/3
 switchport mode access
 switchport access vlan <DATA ID>
 exit
interface fa0/4
 switchport mode access
 switchport access vlan <DATA ID>
 exit
```

Ports	Required assignment
Fa0/2	Data
Fa0/3	Data
Fa0/4	Data

Trunk and unused ports

```
interface fa0/1
 switchport mode trunk
 switchport trunk allowed vlan <MGMT>,<RECEP>,<DATA>
 switchport trunk native vlan <NATIVE ID>
 exit
interface range fa0/5 - 8
 switchport mode access
 switchport access vlan <DEAD ID>
 shutdown
 exit
```

- AN-2 Fa0/1 must match the AN-1 trunk allowed list and native VLAN.
- AN-2 Fa0/5 through Fa0/8 must be assigned to Dead_Vlan and shut down.

Layer 2 verification and save

```
show vlan brief
show interfaces trunk
copy running-config startup-config
```

Switch	Required actions	Instructor checks
AN-1	Run both show commands and save.	Five VLANs, correct port membership, Dead_Vlan ports down, both trunks correct, configuration saved.
AN-2	Run both show commands and save.	Five VLANs, correct port membership, Dead_Vlan ports down, matching trunk, configuration saved.

The instructor should require the student to read the output against the Binder, not merely execute the show commands.

Not part of the Part 2 Layer 2 check

These commands may exist in the broader simulator, but they are not required for the Layer 2 task described in the guide:

```
ip address
no ip address
ip route
no ip route
encapsulation dot1q
switchport trunk encapsulation dot1q
show ip route
show ip interface brief
show interfaces
show mac address-table
ping
enable password
```

Current implementation watch points

- The required command syntax is mostly present in the current processor.
- show vlan brief currently reports VLANs as active even when Dead_Vlan was suspended.
- show vlan brief does not clearly expose that Dead_Vlan ports are shut down.
- show interfaces trunk does not fully distinguish configured trunks from operationally active trunks.
- The current CLI accepts extra Layer 3 commands on switches, but those commands should not expand the Layer 2 evaluation scope.

Source

Net Capstone Instructor Guide (2), Part 2 - Layer 2, pages 9-15. Reviewed October 20 2026.